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Una variedad algebraica afín es irreducible si y solo si su anillo de coordenadas es un dominio de integridad

Proposición 1. *Sea $X \subset \mathbb{A}^n_K$ una variedad algebraica afín, entonces*

$$X \text{ es irreducible} \iff K[X] \text{ es un dominio de integridad.}$$

***Demostración*:** Por la `prop-variedad-algebraica-afin-irreducible-iff-ideal-primo/`??, X es irreducible $\iff$ $\mathbb{I}(X)$ es un ideal primo y por la `prop-ideal-primo-iff-cociente-di-integridad` $\mathbb{I}(X)$ es un ideal primo $\iff$ $K[x_1,\dots,x_n]/\mathbb{I}(X)$ es un dominio de integridad. Luego por definición de $K[X]$, se tiene el resultado. ■

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